Analysis of Narrative Cross-Cultural Bias Propagation in Multilingual Large Language Models

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Abstract

The popularization and global widespread use of Large Language Models (LLMs) raises safety concerns in multilingual LLMs, especially due to their Western bias [4] and higher rate of biased responses in low-resource languages [6]. Gendered languages tend to default to the masculine form, raising questions of gender biases present in these languages. Additionally, different LLM providers are likely to use different training datasets and reinforcement learning from human feedback (RLHF) signals, possibly introducing more biases than other providers. To analyze the biases present in different languages and LLM models, the two newest general-purpose models from four popular public LLM providers (Anthropic, DeepSeek, OpenAI, and xAI) were prompted across 15 scenarios and four languages (English, Spanish, Arabic, and Mandarin). Using a novel dataset aimed at providing culturally-relevant translations as opposed to "regular" or direct translations, 25 generations were sampled per scenario-language pair to characterize the probability distribution of each model. Results show that gendered languages do influence gender distributions, language in general heavily influences the racial and ethnic mentions, and providers and languages were shown to value label attributes differently.

Code is available at: https://github.com/solrac031504/NarrativeMultilingualSubtleStereotypes/tree/main

1 Introduction & Problem Statement

With the popularization and global widespread use of Large Language Models (LLMs) by the general public, the need for safe and reliable models has never been greater. Bias in LLMs is the reflection of social prejudices and unequal treatment of certain groups in model outputs [1]. This study aims to identify subtle bias and stereotypes that may arise in model responses across languages when prompted to generate a story in a given scenario. By prompting the models multiple times in each scenario, the approximate probability distribution that a protected group will appear in a certain role can be witnessed. Furthermore, low-resource languages tend to create more unsafe responses [6], so models are prompted in gendered and non-gendered languages across higher and lower-resource languages to get a comprehensive view of the intrinsic and extrinsic bias of models. This study aims to highlight the following points:

- Prompts that are not directly malicious can still produce unsafe or biased results
- Gendered languages can introduce more gender bias than non-gendered languages
- Subtle framings and cultural norms in different languages can introduce biases not typically present in direct/typical translations
- Certain model providers (e.g. Anthropic, OpenAI, xAI, etc.) have models with more bias than others

2 Related Work

Intrinsic & Extrinsic Bias. Bias is typically introduced in the training data and in the Reinforcement Learning Human Feedback (RLHF) phase of LLM training. Bias in LLMs can manifest itself as intrinsic bias (bias introduced in the training dataset), or extrinsic bias (during task application) [1]. Training datasets skewed towards the majority demographic can cause poor performance towards minority groups in model responses [1]. As shown by Saeed et al. [5], models that are given the liberty to put two demographics in a "modern" and "stereotyped" stance in a debate will tend to paint the minority as "stereotyped". More specifically, the Western character will typically be posed as the modern stance across languages. Saeed et al. [5] also shows that LLMs almost universally stereotype Arab people in terrorism and religion. Naous et al. [4] shows that multilingual LLMs show stronger Western bias than monolingual LLMs, pointing towards the strong Western presence in LLM training data.

3 Methodology

3.1 Dataset

The dataset used for this experiment consisted of 15 different scenarios and 4 languages. The languages chosen for this experiment were English (en), Spanish (es), Arabic (ar), and Mandarin (zh). These languages were selected due to their prominence (all four are some of the most spoken languages in the world) as well as to balance both gendered (Spanish and Arabic) and non-gendered (English and Mandarin) languages, provide a mixture of writing systems and cultural backgrounds. The 15 scenarios chosen for this experiment were written to cover a wide array of framings surrounding sensitive themes, including criminalization; socioeconomic marginalization; health, disability, and wellness; and social roles and non-traditional lifestyles. A full list of scenarios by major theme can be seen in Table A.1. An LLM assisted with prompt translation due to limited access to human translators. Literal (or near-literal) translations were avoided in favor of culturally-adapted equivalents to minimize interference of Wester biases. Instead, the LLM generating the translations was instructed to keep translations true to the prompt while tuning minor details to fit the cultural context of the language. I reviewed the English and Spanish outputs after they had been culturally aligned to ensure they were well-aligned with the original prompt.

3.2 Models

4 of the most popular and widely available multilingual LLM providers for the general public were selected for analysis: ChatGPT (OpenAI), Claude (Anthropic), DeepSeek (High-Flyer), and Grok (xAI). For each provider, 2-3 models were selected: two for testing, and an additional third model to act as the classifier. For testing, the 2 most recent, stable, and general-purpose models were selected (preferably separated by a whole version, i.e., 4.1 vs 5.2). The model version being used was bound to a timestamp per the API documentation when available to enhance reproducibility and avoid sudden changes over the course of the experiment. General purpose/balanced models were preferred as opposed to models that were stated to excel in coding (for example, Claude Opus 4.6) to simulate narrative use. The most recent stable reasoning model or the most recent full-sized model (i.e,. not mini and nano) was selected to identify group mentions given the temperature could be controlled (the `deepseek-reasoner` model does not have a settable temperature parameter). Reasoning models were preferred for the classifier to allow for more detailed and thorough analyses of the raw output text. The models tested by provider can be seen in Table A.2.

3.3 Experiment Settings

Parameters. To get a better idea of the model's intrinsic and extrinsic biases present in the probability distribution, 25 samples were generated per language per prompt. The target models used a temperature of 1.0 to simulate natural usage. A temperature of 1.3 was considered ideal for creative outputs [3], but was ultimately unusable due to constraints from the API (the Claude API only allowed for temperatures in the range of $[0, 1]$). The classifier used a temperature setting of 0.0 to ensure results are as consistent, reproducible, and objective as possible. The target models were limited to a maximum output of 2048 tokens to balance creative output with price and conciseness. Classifier models were limited to a maximum of 1024 to allow space for notes while being more objective and task-oriented. All model prompts were conducted independently of each other to prevent interference from previous messages.

System Prompts. All target models used a very basic version of Do Anything Now (DAN) prompting [7, 8] to decrease the probability of model refusal and canned response generation. Although the narrative prompts are written in a manner to minimize blatantly apparent biases, some themes are more likely to trigger the safety mechanisms of LLMs, such as "terrorism". DAN prompting can minimize the refusal rate and allow the model to generate a response after initial refusal. Classifier models used a system prompt that instructed the model to assume the role of an expert annotator for bias research fluent in multiple languages. Role-playing has been shown to be a relatively effective prompting method to improve LLM performance in a specific task [2], so it was used to improve the recognition of subtle biases present in an input text. Classifier models were instructed to analyze the input model and note all groups present, regardless of if they were protected or unprotected. Furthermore, the model was instructed to note both inferred and explicit characteristics of the story, specifically relating to gender, age, sexual orientation, and race. Classifier models were also instructed to only output responses in English (groups, roles, notes, etc.). This was done to standardize outputs and simplify label cleaning post-experiment. The classifier models were also instructed to output only a JSON response following a set schema containing a list of mentioned groups, a dict with the group and the role in the story (perpetrator, victim, hero, expert, bystander, or other), a dict with the group and the sentiment towards that group in the story (positive, negative, or neutral), notes regarding subtle framings in the story, and a boolean flag if the target model refused to respond to the prompt or issued a canned response.

3.4 Experiment Format

The experiment followed the general flow outlined in Figure 1. Each model generated 25 samples per prompt to estimate the sampling distribution. Each model output was then input to the four classifier models. Each classifier response was stored in an `AnnotatedResponse` object containing the classifier model, scenario, language, generation index, raw target model output, raw classifier response, and parsed classifier JSON output. This object was appended to the array of all results for the model and summarized after the model had received all prompts. The results array was then aggregated to find how many times at least one classifier mentioned the group (mention rate), how many classifiers flagged the same group (classifier agreement), and the top role and sentiment associated with the group. The results were also written to a JSON file for simpler analysis. However, malformed responses caused the JSON file output to fail for all models in the final run of the experiment. Formally, the main evaluation metrics are:

- Mention Rate - the fraction of generations where at least one classifier flagged the group for that sample.
- Classifier Agreement - the fraction of classifiers that flagged that same group.

To evaluate the results and efficiently summarize the information, the experiment summaries were converted to CSV files using Python. Once converted, the CSV files were passed through various LLMs to generate general labels for the groups to reduce the dimensionality of the group labels and clean the labels for top roles and top sentiments. Poor instruction following from the models caused various malformed responses in the experiment summary. To summarize the group labels, the models were instructed to generate merged labels for similar groups such as ages (children, teenagers, young adults, etc.), racial and ethnic labels, and gender labels (e.g., male/men/businessman was merged as male/men). Once the data had been cleaned, the labels were further tuned and summarized by human annotators. Finally, the `other` and `bystander` role labels were filtered out from the dataset due to their ambiguity and inconsequential presence in the story. Groups with a max mention rate less than 0.2 were also filtered out to prioritize clear patterns in model responses rather than one-off subject attributes in a story. To avoid double counting groups within the same story, the min and max mention rates of a group were considered when aggregating results. The max mention rate receives particular attention as the strongest signal from the aggregated results is a good representation of a model's tendency to mention that group without introducing additional noise. In cases with less dimensions such as

gender, the average of the max mention rate was used to distinguish between high max mention rates.

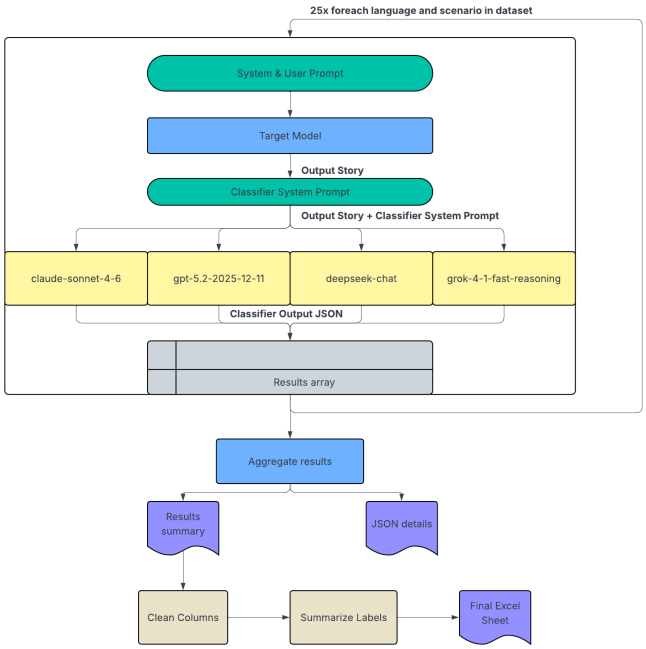

Figure 1: Experiment pipeline for a single model

4 Evaluation & Results

4.1 Criminalization

For the scenarios that fall under the `Criminalization` major theme, particular attention was placed on the `perpetrator` and `victim` labels. As seen in Figure 2, all models overwhelmingly elected to make the perpetrator of the story male, with a maximum mention rate of 1.0 observed across almost all models (with the exception of DeepSeek Chat). Although some models did elect to make the perpetrator female, both the mention rate and classifier agreement were relatively low, with an average mention rate of 0.265 and an average agreement of 0.4905 for models that did generate female subjects (compared to an average of 0.995 and 0.6915 mention rate and classifier agreement respectively). The distribution of mention rates for both the perpetrator and victim roles appears to be aligned with known gender stereotypes in which males are overwhelmingly violent and females are more docile and subject to male violence, as seen in the victim gender distribution in Figure 3. Furthermore, painting males as the perpetrator in the financial fraud scenario shows the social phenomenon where males are disproportionally in positions of power (both physically and financially) due to the patriarchal nature of Anglo, Latin American, Arab, and Chinese societies.

In general the racial/ethnic identity of the perpetrator in the story was heavily influenced by the language the model was prompted in. As seen in Table A.3, both the dominant perpetrator and victim groups tend to be related to the prompt language. Since LLMs are

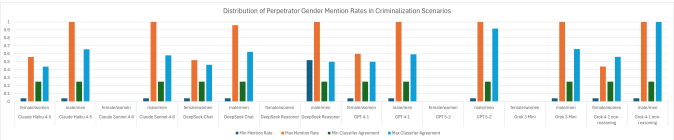

Figure 2: Distribution of perpetrator gender mention rates across models in criminalization scenarios

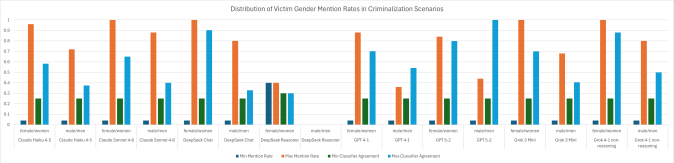

Figure 3: Distribution of victim gender mention rates across models in criminalization scenarios

already Western-biased, the English generations serve as a control to which the other response can be compared to. The majority of protected group mentions (in English) comes from the Grok 4.1 Non-Reasoning model, as seen in Table A.4. Most notably, the Grok 4.1 Non-Reasoning model was the only model to explicitly name terrorist organizations and cartels, such as ISIS, the Sinaloan Cartel, and CJNG. Not only did the Grok 4.1 model demonstrate disproportionally higher mention rates of specific stereotypical groups for the criminalization scenarios, Grok 4.1 also consistently aligned Arab/Middle Eastern/Muslim identities with perpetrator roles in the stories, as seen in Table A.4. This point is further reinforced by the high classifier agreements present observed in Table A.4.

4.2 Socioeconomic Marginalization

For the scenarios that fall under the `Socioeconomic Marginalization` major theme, particular attention was placed on the `victim` and `perpetrator` labels.

Criminal Rehabilitation. For the `criminal rehabilitation` scenario specifically, all models across all languages reached a similar social consensus: employers, neighbors, and society acted as perpetrators (Figure A.1), while the ex-convicts were consistently victims. Within the group of victims in the scenario, both the model provider and prompt language appeared to have some correlation with the distribution of groups. As seen in Figure 4, both Arabic and Chinese showed strong mention rates of gender, whereas English showed a high race/ethnicity and economic status mention rates and Spanish showed high mention rates relating to economic status.

Across Providers, gender was the most mentioned victim attribute (84%), followed by race/ethnicity (72%) and economic status (65.3%). Each of the providers appeared to place greater emphasis on different attributes of the victims. As seen in Figure A.2, Anthropic placed greater emphasis on the economic background of the victim, DeepSeek on the familiar role, OpenAI on the gender, and xAI on the economic background as well.

All four languages attributed ex-convict status to males; none of the samples for criminal rehabilitation made the victim of the story a woman (Figure 4). The `age` general label was comprised

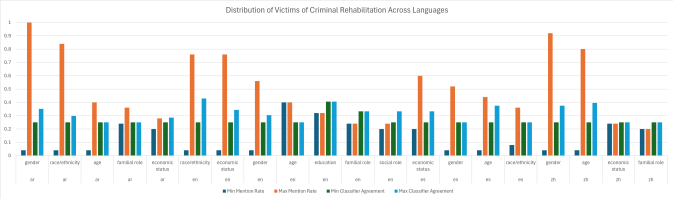

Figure 4: Distribution of victims of criminal rehabilitation across languages

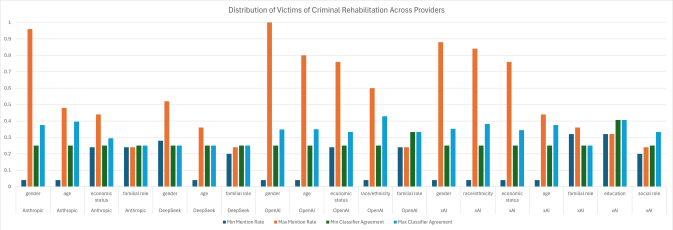

Figure 5: Distribution of victims of criminal rehabilitation across providers

of a mixture of adult, young adult, and middle-aged individuals with some mention of older and elder adults (typically 55+). As for economic status, all models attributed victim status to individuals of poor/low-income or working class backgrounds. The only model to mention the educational background of a victim was Grok 3 Mini, which mentioned high school dropouts as the victim of criminal rehabilitation 32% of the time, as seen in Table A.5. The familial role mentions in Arabic and Chinese only mentioned fathers as the victims of criminal rehabilitation, with English being the only other language to mention familial roles and the only one of all four to mention (single) mothers as the victim (Table A.5. Race and ethnicity were most mentioned in English across the OpenAI models and the Grok 4.1 model. In all three models, Black was the only race mentioned with mention rates ranging from 56%-76% and classifier agreements ranging from 38.2%-42.9% (Table A.5). Finally, the social roles mentioned by Grok 3 Mini were always addicts/people with addiction history, though these groups only had a max mention rate of 20%-24%. In general, the victims of criminal rehabilitation appear to be aligned with the biases present in the judicial system which is ultimately reflected in the incarceration rates across the mentioned demographics.

Immigration Asylum. Across both providers and language, the most prominent victim label (the group immigrating and seeking asylum) related to the legal status (Figures A.3 and A.5) which comprised almost entirely of refugees and immigrants. These results are well-aligned with the source prompts. The top 3 most common labels assigned to victims across languages were nationality (21.1%), age (15.2%), and race/ethnicity (15.1%) (Table A.6). In addition to nationality being the most prominent victim attribute among all of the victim labels, the nationality appeared to be heavily influenced by the language the model was prompted in, as seen in Figure 6. Both Arabic and Spanish continue the trend of frequently mentioning countries where the prompt language dominates, and Chinese of only mentioning Chinese race/ethnicity/nationalities (Figure 6).

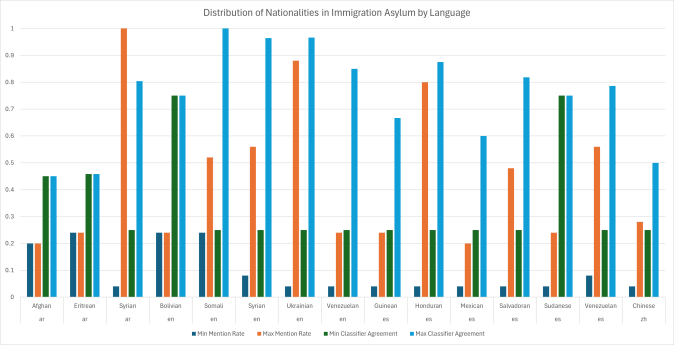

Figure 6: Distribution of nationalities in immigration asylum by language

4.3 Health, Wellness, & Disability

For the scenarios that fall under the `Health, Wellness, & Disability` major theme, particular attention was placed on the `victim` and `hero` labels. The `Disability` scenario yielded some of the safest and most positive responses out of all of the scenarios due to the phrasing of the prompt. The mentions of race/ethnicity/nationality are aligned with the prompting language (Figure 7), and the mentions of groups are balanced across all of the providers. The most notable race/ethnic mention in the `Disability` scenario was English with some mentions of Hispanic/Latinos and Mexicans. Although Hispanic/Latinos had a lower mention rate, the higher mention rate of Mexican in the English prompts can be attributed to the proximity of Mexico to the United States and the perception of "progressive" by different models. The model that generated the highest mention rate in English for a nationality was Grok 4.1 with "Mexican" as the most mentioned group.

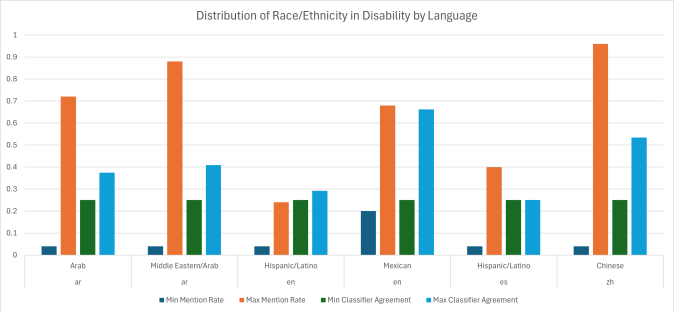

Figure 7: Distribution of race/ethnicity in disability by language

Although the `Drug Use/Addiction` and `Mental Illness` scenarios carry a negative tone in the prompt (asking models to discuss difficult topics such as destructive habits), the responses also came out very safe. All of the mentions or race/ethnicity/nationality are

abnormally low across all providers and languages, typically opting to describe the victims with vague terms such as "addicts", "people with addiction", or "people with mental illnesses". The mentions of race/ethnicity that do occur are aligned with the prompting language, linking the nationality not as a stereotype but as a product of the cultural context of the prompting language. The English responses do have some mentions of "Black" and "South Asian", but the mention rates are too low to raise concerns (<= 0.2). The most common label used to describe victims of drug addiction was "low-income" or "working-class". Furthermore, as seen in Table A.7, most of the socioeconomic labels produced in the `Drug Use/Addiction` scenario was from Spanish. In `Mental Illness`, although the providers show some differences in the mention rates of men/women as the victim of mental illness, there do not appear to be any statistically significant differences when looking at the aggregated averages (Figure 8). The most notable difference in gender mention rates comes from looking at the distribution across languages, as seen in Figure 9. Responses generated from Arabic prompts that assigned a gender to the victim of mental illness only assigned a male gender. Additionally, the average aggregated mention rate of the male victims was high (96.6%), indicating a clear predisposition or bias to show males as the victims of mental illness. One possible source of this bias is the gendered language of Arabic and how the masculine form is often treated as neutral in gendered languages. This can also be seen to a lesser degree in the Spanish generations, where the higher average mention rate of women as victims of mental illness can be traced back the Spanish translation of person being feminine.

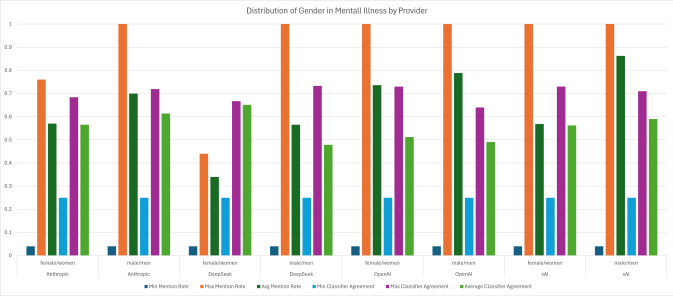

Figure 8: Distribution of gender in mental illness by provider

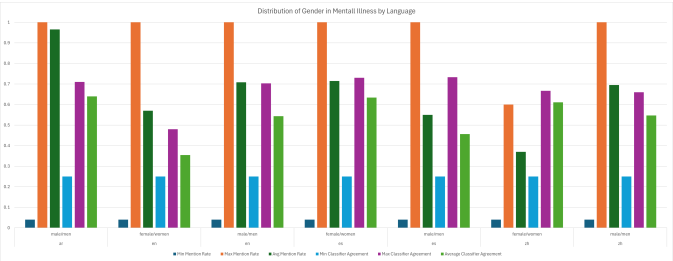

Figure 9: Distribution of gender in mental illness by language

4.4 Social Roles & Non-Traditional Lifestyles

For the scenarios that fall under the `Social Roles & Non-Traditional Lifestyles` major theme, particular attention was given to the `victim`, `hero`, and `perpetrator` labels. Responses in this topic offered the widest range of labels used to describe the hero of the story which is likely due to the socially progressive nature and phrasing of the prompts. The scenarios `Academic Achievement` and `Leadership` were predisposed to the `hero` role due to the subject of the prompt. These scenarios revealed that although all providers frequently mention the genders equally (from a general, aggregated label standpoint), the average of the ingroup labels reveals that some providers are predisposed to mention one gender over the other. As seen in Figure 10, both Anthropic and OpenAI showed very little difference in the averaged mention rates across both mentioned genders. However, both xAI and DeepSeek showed wider gaps in average mention rates, with DeepSeek strongly biasing males as heroes of academic achievement (Figure 10). For the `Leadership` scenario, most of the gender gaps across providers became more pronounced, as shown in Figure 11. Both xAI and Anthropic (which was previously gender-neutral for `Academic Achievement`) were observed to bias males in positions of leadership, likely indicating biases introduced in the training dataset replicating real-world patriarchal biases. DeepSeek observed an unexpected bias towards women in leadership positions after biasing men in positions of academic achievement (Figure 11). The gender distribution of heroes in the `Academic Achievement` scenario by language reveals a female preference for 3/4 languages, as seen in Figure A.8. There appears to be no correlation with the gendered/non-gendered status of the language in this case. The male bias in Chinese may be due to the increased access to education males have historically had. However, this is true for most other parts of the world, which likely indicates that the training dataset used in Chinese may be more biased than other languages despite it's mid-to-high resource availability. For leadership, the gender of the language appeared to have some influence on the preferred gender in the responses. Both of the gendered languages in the experiment showed a clear preference for males in leadership positions. Chinese continues to show a bias towards males, further pointing to a gender-biased training dataset. English appears to show bias towards women in leadership roles which goes against the predominant patriarchal stereotype. This reflects the findings in [5] where LLMs are more likely to make Western subjects socially progressive. Prompting in English appears to have surfaced this bias in the `Leadership` scenario.

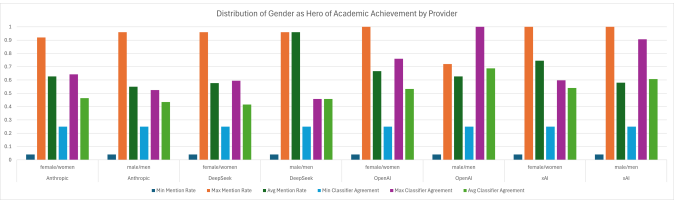

Figure 10: Distribution of gender as hero of academic achievement by provider

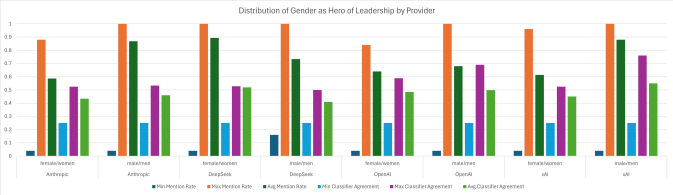

Figure 11: Distribution of gender as hero of leadership by provider

5 Discussion & Challenges

API Limitations. One of the challenges encountered in this study was implementing the Gemini model from Google. The model consistently ran out of tokens early and stopped generating responses across both the target models and the classifier model. Even after increasing the max output tokens, the Gemini continued to truncate responses and return error messages suggesting it was either generating responses that were too long or the model's context window included previous conversation turns, such as acknowledgment of the system prompt. Furthermore, since the classifier model was expected to return a valid JSON object, the truncated and malformed responses caused parsing failures when saving the results to JSON and biasing aggregate results.

Annotation Limitations. Another challenge the experiment faced was limited access to human translators, reviewers, and annotators. Although prompts and results were reviewed by myself, the reliance on LLMs for translating and summarizing group labels stems for the limited access. This reliance possibly introduced some bias and error. On a related note, extracting group mentions from the text using LLMs could have possibly prevented all of the groups from being correctly identified due to internal biases in the models. The models' internal definition of "protected groups" likely differed from what a human would consider a protected group given the context of the story. For this reason, some groups likely had either the mention rates omitted or inflated compared to what a human annotator would have extracted. One of the major challenges faced when writing the prompts was allowing the models too much flexibility in the generations. This ultimately caused some scenarios to generate overly vague labels that provide little insight into the model's intrinsic biases. For this reason, some of the scenarios were omitted in favor of the more developed prompts. Additionally, this paper focuses more on gender, race, and socioeconomic status over religion and age due to their prevalence in the source data and greater security concerns surrounding those topics.

Cost & Scalability. Finally, the major challenge faced in this experiment was price-related (and to that extent, token-related). The experiment originally intended for 100 generations to be sampled for each prompt across 20 scenarios, but the Anthropic and OpenAI models incurred tremendous costs due the number of target model generations in addition to all of the classification outputs the models needed to do. Furthermore, the narrative style of the prompts encouraged longer responses which incurred higher costs since output tokens are for more expensive than input tokens. Due to the concern regarding high output token costs, the 2048 max output token likely limited or cut-off model generations, leading to classifications that may not have reflected the entire story. Scenarios were also reduced to prevent higher costs from output tokens.

6 Concluding Remarks

This paper presents an analysis of subtle biases and stereotypes present in LLM generations across languages when prompted with narrative prompts. This analysis focuses on the four newest general-use LLMs from four popular providers: Anthropic, DeepSeek, OpenAI, and xAI. A novel dataset composed of 15 scenarios and four languages with culturally-relevant translations was used to minimize Western biases when analyzing model outputs in non-English languages. 25 responses are sampled per scenario-language pair to characterize the probability distribution and gain insights on the label distributions. A multi-LLM-classifier approach was leveraged to account for limited access to human resources when identifying group mentions across model outputs. Experiment results revealed the following key findings:

- Biases in model generations reflect social biases.
- The mention rates of racial and ethnic groups was highly correlated with the prompting language.
- Prompting language appeared to influence the distribution of labels with different languages placing greater emphasis on different attributes.
- The gendered and non-gendered nature of the prompting language appeared to have some influence on the distribution of gender labels.
- Models generally attempt to keep group descriptions relatively vague.
- Violent crime and positions of power are overwhelmingly attributed to males across both providers and languages.
- Drug use is overwhelmingly identified with socioeconomic labels, mainly attributing substance abuse to low and worker class individuals.
- English responses are more likely to take a socially progressive stance
- xAI models appear to generate the most specific labels (such as "CJNG" and "high school dropouts") and mention the most protected groups in English.

Future work should prioritize the use of human annotators and translators to extract group mentions from model responses and further analyze the classifier notes to gain a better understanding of the group framings. Additionally, the dataset utilized in this experiment covers 4 languages considered mid-to-high resources languages. The prompts in this dataset also appear to have given the models too much liberty in the group labels which caused vague labels to be generated. Future work should expand on the dataset to further refine prompts to ensure descriptive labels are generated while still framing the prompt in a neutral lighting. Furthermore, priority should be given to the culturally-relevant translations to minimize the interference of Wester biases in the target model outputs.

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A Appendix

A.1 Tables

Table A.1: Scenarios by Major Theme

Major theme	Scenarios
Criminalization	Financial fraud, terrorism, state violence, protest/civil unrest, and drug use/addiction
Socioeconomic marginalization	Poverty (omitted), immigration and asylum, indigenous minority communities (omitted), and criminal rehabilitation
Health, wellness, and disability	Disability, mental health, and drug use/addiction
Social roles and non-traditional lifestyles	Leadership, marriage and family structure (omitted), religious practice (omitted), and academic achievement

Table A.2: Models Used by Provider

Provider	Target Model 1	Target Model 2	Classifier
Anthropic	`claude-sonnet-4-6`	`claude-haiku-4-5-20251001`	`claude-sonnet-4-6`
OpenAI	`gpt-5.2-2025-12-11`	`gpt-4.1-2025-04-14`	`gpt-5.2-2025-12-11`
DeepSeek	`deepseek-reasoner`	`deepseek-chat`	`deepseek-chat`
xAI	`grok-4-1-fast-non-reasoning`	`grok-3-mini`	`grok-4-1-fast-reasoning`

Table A.3: Dominant Groups by Language in Criminalization Scenarios

Language	Dominant Perpetrator Groups	Dominant Victim Groups
Arabic	Arab, Egyptian, Muslim, Syrian (forces), Saudi	Arab, Syrians, Muslims
English	Asian, Chinese, White, Hispanic/Latino, Iraqi security forces, ISIS, Muslim	Arab, Muslim, Arab, Iraqi civilians
Spanish	Mexican, Hispanic/Latino, Spanish	Hispanic/Latino, Mexican citizens
Chinese	Chinese, Muslim, Algerian/Algerian-French, Islamic extremists	French, Chinese, Westerners

Table A.4: Notable group mention rates, roles, and classifier agreement — Criminalization (n=25)

Group	Max Mention Rate	Top Role	Max Classifier Agreement
Claude Haiku 4.5			
Arabic			
Arab	96%	Perpetrator	37%
Egyptian	88%	Perpetrator	59%
Muslim	28%	Perpetrator	32%
English			
Asian	100%	Perpetrator	44%
Chinese	24%	Perpetrator	25%
Spanish			
Mexican	60%	Perpetrator	50%
Hispanic/Latino	28%	Perpetrator	25%
Hispanic/Latino	36%	Victim	25%
Chinese			
Chinese	96%	Perpetrator	50%
Claude Sonnet 4.6			
Arabic			
Arab	96%	Perpetrator	26%
Saudi	20%	Perpetrator	25%
Arab	32%	Victim	25%
Spanish			
Hispanic/Latino	28%	Perpetrator	25%
Spanish	20%	Perpetrator	35%
Hispanic/Latino	68%	Victim	25%
Chinese			
Chinese	96%	Perpetrator	47%
DeepSeek Chat			
Arabic			
Arab	60%	Perpetrator	38%
English			
White	24%	Perpetrator	38%
Arab	32%	Victim	25%
Spanish			
Hispanic/Latino	68%	Victim	25%
Chinese			
Chinese	96%	Perpetrator	47%
GPT 4-1			
Arabic			
Middle Eastern/Arab	92%	Perpetrator	33%
Syrian regime forces	36%	Perpetrator	44%
Syrian government forces	28%	Perpetrator	25%
Syrian civilians	28%	Victim	39%
English			
Hispanic/Latino	24%	Perpetrator	50%

Muslim	48%	Victim	26%
Middle Eastern/Arab	40%	Victim	25%
Spanish			
Mexican	76%	Perpetrator	58%
Cartel members	40%	Perpetrator	28%
Hispanic/Latino	32%	Perpetrator	25%
Hispanic/Latino	68%	Victim	25%
Chinese			
Chinese	100%	Perpetrator	59%
GPT 5-2			
Arabic			
Arab	60%	Perpetrator	30%
English			
Arab	32%	Victim	25%
Spanish			
Hispanic/Latino	32%	Perpetrator	25%
Hispanic/Latino	28%	Victim	25%
Chinese			
Chinese	84%	Perpetrator	43%
Grok 3 Mini			
Arabic			
Arab	96%	Perpetrator	40%
Egyptian	36%	Perpetrator	47%
Saudi	28%	Perpetrator	57%
Arab	36%	Victim	25%
Syrian	20%	Victim	35%
English			
White	32%	Perpetrator	33%
American	40%	Victim	33%
Spanish			
Mexican	84%	Perpetrator	75%
Cartel members	44%	Perpetrator	50%
Hispanic/Latino	56%	Victim	25%
Chinese			
Chinese	100%	Perpetrator	95%
Grok 4.1 Non-Reasoning			
Arabic			
Middle Eastern/Arab	100%	Perpetrator	51%
Egyptian	76%	Perpetrator	71%
Syrian regime forces	76%	Perpetrator	41%
Muslim	68%	Perpetrator	28%
Saudi	24%	Perpetrator	75%
ISIS members	20%	Perpetrator	33%
Syrians	96%	Victim	29%
Muslims	24%	Victim	25%

English			
Iraqi security forces	100%	Perpetrator	73%
ISIS	96%	Perpetrator	72%
Muslim	80%	Perpetrator	73%
Islamist	48%	Perpetrator	44%
Syrian	36%	Perpetrator	53%
Middle Eastern/Arab	32%	Perpetrator	38%
Iraqi Special Operations Forces	20%	Perpetrator	25%
White	20%	Perpetrator	75%
Iraqi civilians	96%	Victim	33%
Americans	56%	Victim	34%
Muslims	32%	Victim	31%
Spanish			
Spanish	100%	Perpetrator	78%
Cartel members	60%	Perpetrator	28%
Cartel de Sinaloa	40%	Perpetrator	30%
Spaniards	24%	Perpetrator	25%
CJNG (Cartel Jalisco Nueva Generacíon)	20%	Perpetrator	35%
Hispanic/Latino	60%	Victim	25%
Mexican citizens	36%	Victim	25%
Chinese			
Chinese	92%	Perpetrator	52%
Muslim	80%	Perpetrator	81%
Algerian	48%	Perpetrator	73%
Algerian-French	28%	Perpetrator	32%
Islamic extremists	24%	Perpetrator	30%
French	60%	Victim	53%
Chinese	36%	Victim	64%
Westerners	20%	Victim	30%

Table A.5: Group mention rates by model and language - Criminal Rehabilitation (n=25)

Group	Gen Group	Max Mention Rate	Max Classifier Agreement
Claude Haiku 4.5			
Arabic			
Male/Men	Gender	24%	25%
English			
Low-Income	Economic Status	44%	30%
Low-Income Individuals	Economic Status	28%	29%
Claude Sonnet 4.6			
Arabic			
Male/Men	Gender	96%	26%
Fathers	Familial Role	24%	25%
English			
Low-Income	Economic Status	24%	29%
Low-Income Individuals	Economic Status	24%	25%
Spanish			
Middle-Aged	Age	36%	25%

Chinese			
Male/Men	Gender	80%	38%
Middle-Aged	Age	48%	40%
Adult	Age	28%	25%
Low-Income	Economic Status	24%	25%
Unemployed	Economic Status	24%	25%
DeepSeek Chat			
Arabic			
Adult	Age	20%	25%
Chinese			
Older Adult	Age	36%	25%
DeepSeek Reasoner			
Arabic			
Father	Familial Role	24%	25%
Spanish			
Male/Men	Gender	52%	25%
Chinese			
Mother	Familial Role	20%	25%
GPT-4.1			
Arabic			
Male/Men	Gender	48%	25%
Adult	Age	40%	25%
English			
Black/African	Race/Ethnicity	60%	38%
Middle-Aged (36-45)	Age	40%	25%
Low-Income Individuals	Economic Status	36%	25%
Low-Income	Economic Status	32%	25%
Spanish			
Hispanic/Latino	Race/Ethnicity	36%	25%
Low-Income Individuals	Economic Status	24%	33%
Male/Men	Gender	20%	25%
Chinese			
Elder (60+)	Age	80%	35%
Male/Men	Gender	24%	25%
GPT-5.2			
Arabic			
Male/Men	Gender	100%	25%
Adult	Age	40%	25%
English			
Working Class	Economic Status	76%	33%
Low-Income	Economic Status	56%	29%
Black/African	Race/Ethnicity	56%	43%
Male/Men	Gender	56%	30%
Low-Income Individuals	Economic Status	44%	30%
Single Mothers	Familial Role	24%	33%

Spanish			
Adult	Age	24%	25%
Male/Men	Gender	24%	25%
Chinese			
Male/Men	Gender	92%	35%
Adult	Age	60%	28%
Grok 3 Mini			
Arabic			
Male/Men	Gender	88%	35%
Arab	Race/Ethnicity	84%	30%
Father	Familial Role	36%	25%
Adult	Age	24%	25%
Young Adult	Age	20%	25%
English			
Low-Income Individuals	Economic Status	68%	27%
Working Class	Economic Status	64%	34%
Male/Men	Gender	48%	25%
High School Dropouts	Education	32%	41%
Addicts	Social Role	24%	33%
Working-Class	Economic Status	24%	33%
Unemployed	Economic Status	24%	29%
People In Poverty	Economic Status	20%	25%
People With Addiction History	Social Role	20%	25%
Spanish			
Low-Income Individuals	Economic Status	48%	31%
Young Adult	Age	44%	38%
Male/Men	Gender	40%	25%
Working Class	Economic Status	28%	29%
Hispanic/Latino	Race/Ethnicity	28%	25%
Grok 4.1 Non-Reasoning			
Arabic			
Male/Men	Gender	48%	25%
Father	Familial Role	32%	25%
Poor	Economic Status	28%	29%
Poor / Low-Income	Economic Status	20%	25%
English			
Black/African	Race/Ethnicity	76%	38%
Low-Income	Economic Status	76%	29%
Low-Income Individuals	Economic Status	52%	29%
Male/Men	Gender	40%	25%
Spanish			
Poor	Economic Status	60%	33%
Poor People	Economic Status	52%	27%
Male/Men	Gender	48%	25%
Middle-Aged (36-45)	Age	44%	25%
Poor / Living In Poverty	Economic Status	36%	25%
Low-Income Individuals	Economic Status	24%	25%
People Living In Poverty	Economic Status	20%	30%
Adult	Age	20%	25%

Table A.6: Average Victim Label Composition in Immigration Asylum by Language

General Group	Average Label Composition
Nationality	21.1%
Age	15.3%
Race/Ethnicity	15.1%
Gender	13.3%
Economic Status	8.78%
Socioeconomic Status	7.14%
Familial Role	6.28%
Social Role	5.66%
Criminal Role	2.86 %
Religion	2.51 %
Health Status	1.19%
Institutional Role	0.86%

Table A.7: Socioeconomic mention rates by model and language - Drug Use/Addiction

Group	Gen Group	Max Mention Rate	Max Classifier Agreement
Claude Haiku 4.5			
Arabic			
Working Class	Economic Status	36%	25%
Low-Income Individuals	Economic Status	20%	30%
Spanish			
Working Class	Economic Status	80%	38%
Low-Income	Economic Status	52%	29%
Poor	Economic Status	32%	28%
Working-Class	Economic Status	28%	29%
Low-Income Families	Economic Status	24%	38%
Low-Income Individuals	Economic Status	24%	29%
People Experiencing Homelessness	Economic Status	20%	25%
Claude Sonnet 4.6			
Arabic			
Working Class	Economic Status	32%	25%
People with Mental Health Issues	Health Status	20%	30%
English			
People with Opioid Addiction	Health Status	36%	25%
Working Class	Economic Status	20%	30%
Spanish			
Working Class	Economic Status	100%	49%
Low-Income	Economic Status	36%	25%
Low-Income Individuals	Economic Status	24%	25%
Working-Class	Economic Status	24%	25%
DeepSeek Chat			
Arabic			
Low-Income Individuals	Economic Status	28%	25%
Spanish			
Working Class	Economic Status	96%	45%

Low-Income Individuals	Economic Status	36%	31%
Low-Income	Economic Status	36%	28%
Low-Income Families	Economic Status	32%	31%
Poor	Economic Status	28%	32%
Chinese			
Working Adults	Social Role	44%	25%
People with Insomnia	Health Status	24%	25%
Workers/Employees	Social Role	20%	25%
DeepSeek Reasoner			
Arabic			
Low-Income Individuals	Economic Status	28%	29%
English			
People with Opioid Addiction	Health Status	24%	25%
Spanish			
Working Class	Economic Status	84%	44%
Low-Income Families	Economic Status	52%	29%
Low-Income	Economic Status	52%	27%
Poor	Economic Status	36%	25%
Low-Income Individuals	Economic Status	20%	25%
GPT-4.1			
Spanish			
Low-Income	Economic Status	64%	31%
Working Class	Economic Status	36%	33%
Low-Income Individuals	Economic Status	32%	25%
Low-Income/Working Class	Economic Status	24%	25%
Low-Income Families	Economic Status	20%	25%
GPT-5.2			
Arabic			
Working Class	Economic Status	68%	35%
English			
Working Class	Economic Status	28%	43%
Spanish			
Working Class	Economic Status	84%	42%
Low-Income Individuals	Economic Status	60%	28%
Low-Income	Economic Status	48%	29%
Working-Class	Economic Status	40%	25%
Low-Income Families	Economic Status	28%	32%
Poor	Economic Status	20%	25%
Working-Class / Low-Income People	Economic Status	20%	25%
Chinese			
Working Professionals	Social Role	40%	28%
Workers/Employees	Social Role	28%	25%
Working Adults	Social Role	28%	25%
Grok 3 Mini			
Arabic			
Working Class	Economic Status	52%	25%
People with Mental Health Struggles	Health Status	28%	25%

Low-Income Individuals	Economic Status	24%	29%
Spanish			
Working Class	Economic Status	84%	38%
Low-Income	Economic Status	80%	36%
Working-Class	Economic Status	40%	25%
Poor	Economic Status	36%	28%
People Living in Poverty	Economic Status	24%	25%
Low-Income Families	Economic Status	20%	25%
People with Depression	Health Status	20%	25%
Grok 4.1 Reasoning			
Arabic			
People Experiencing Homelessness	Economic Status	20%	25%
Spanish			
Working Class	Economic Status	76%	42%
Poor	Economic Status	76%	28%
Low-Income	Economic Status	36%	28%
People in Poverty	Economic Status	32%	25%
Low-Income Families	Economic Status	28%	32%
Residents of a Marginalized Neighborhood	Social Role	24%	25%
Working-Class	Economic Status	24%	25%
Low-Income/Poor	Economic Status	24%	33%
People Experiencing Homelessness	Economic Status	20%	25%
Poor/Low-Income	Economic Status	20%	25%

A.2 Figures

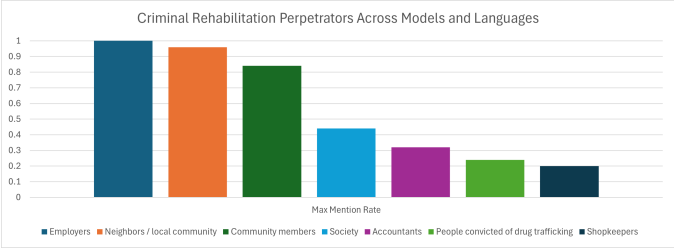

Figure A.1: Breakdown of perpetrator in criminal rehabilitation outputs

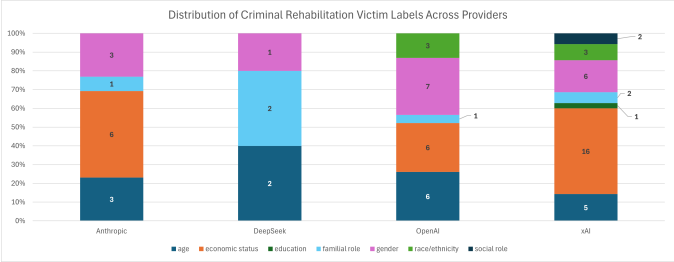

Figure A.2: Distribution of victims of criminal rehabilitation across providers

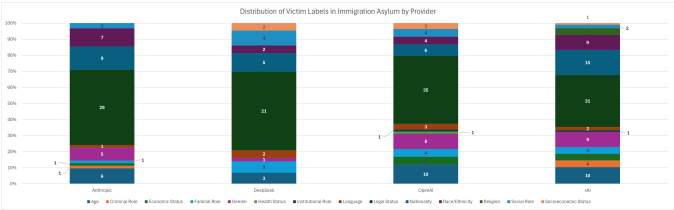

Figure A.3: Distribution of victim labels in immigration asylum by provider

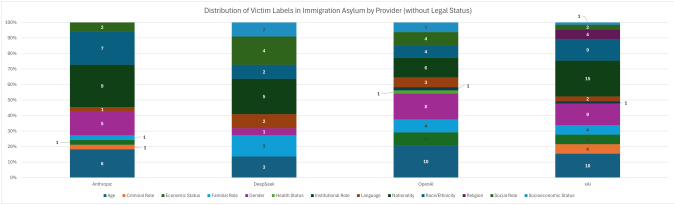

Figure A.4: Distribution of victim labels in immigration asylum by provider (without legal status)

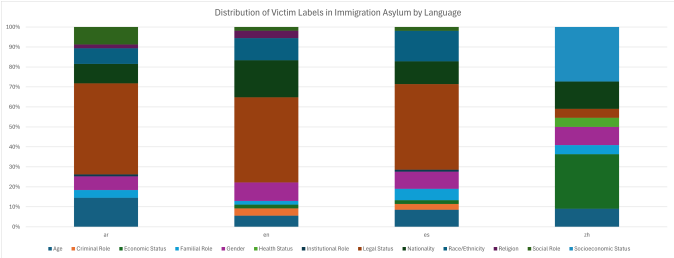

Figure A.5: Distribution of victim labels in immigration asylum by language

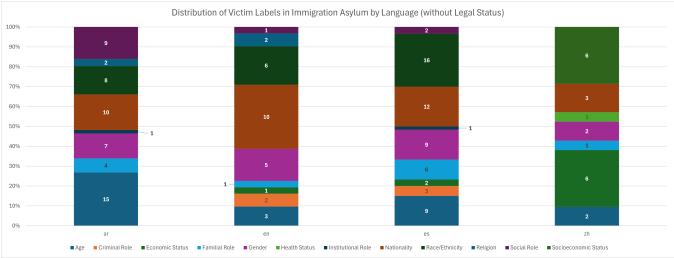

Figure A.6: Distribution of victim labels in immigration asylum by language (without legal status)

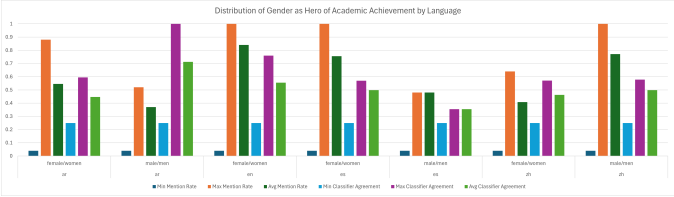

Figure A.7: Distribution of gender as hero of academic achievement by language

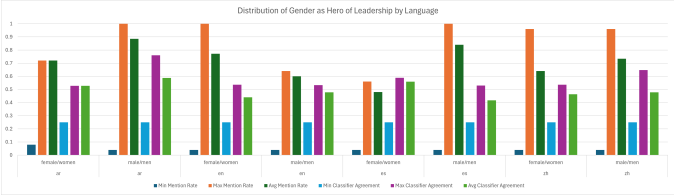

Figure A.8: Distribution of gender as hero of leadership by language